

RUTHISHKUMAR G

+91 7010329076 ruthishkumarg@gmail.com linkedin.com/ruthishkumar18 github.com/ruthishkumar18

PROFESSIONAL SUMMARY

An enthusiastic learner with strong interest in **Artificial Intelligence, Machine Learning, Generative AI, and Full Stack Development**. I enjoy building practical applications and solving real-world problems through hands-on development. Continuously exploring new technologies and improving skills through project-based learning.

EDUCATION

M.Tech – Computer Science and Engineering (Integrated 5-Year Program) Aug 2023 – Present

Sri Ramakrishna Engineering College, Coimbatore, Tamil Nadu

CGPA: 8.38 | Active Member – Association of Master's Tech CSE

Higher Secondary School (HSS) 2023

Government Higher Secondary School, Theni, Tamil Nadu

Percentage: 87% | Academic Merit | Class Rank: 1

Secondary School Leaving Certificate (SSLC) 2021

Government High School, Theni, Tamil Nadu

Result: Pass | State Board of Tamil Nadu

TECHNICAL SKILLS

Programming	Python, C, Java (Basics), JavaScript, HTML5, CSS3
AI / ML & DL	TensorFlow, YOLO, CNN, Prompt Engineering , RAG, LLM-based Applications
Frameworks	ReactJS, Flask, Streamlit, AngularJS (Basics), Node.js (Basics)
Databases	MySQL, MongoDB, SQLite3, Firebase
Tools	Git, GitHub, LM Studio, Colab, VS Code, Roboflow, LabelMe, Android Studio, Jira
Core Concepts	Data Structures, Algorithms, OOPS, API Integration

AREA OF INTEREST: AI / ML & DL, Generative AI, Full Stack Development, LLMs

PROJECTS

PlantSpeakAI – AI Enabled Plant Electrophysiology Monitoring System Oct 2025 – Apr 2026

Tech Stack: ESP32, Python, Random Forest, Neural Network, Signal Processing, Android Studio

- Engineered an intelligent plant monitoring system that captures plant **bio-electrical signals** using ECG electrodes connected to an **ESP32 microcontroller**.
- Extracted signal features and trained **Machine Learning models** such as **Random Forest** and **Neural Networks** to classify plant stress conditions.
- Developed an **Android mobile application** to visualize plant health status and send alerts when abnormal plant stress signals are detected.

JudifyAI – Intelligent Case Law Audio Platform Feb 8 2026 – Mar 1 2026

Tech Stack: Cheerio, Ollama (Mistral), Piper TTS, React.js, JWT Authentication

- Developed an AI-powered legal platform that summarizes lengthy court judgments using **Large Language Models (LLMs)**.
- Implemented **web scraping** using Cheerio and integrated **Piper Text-to-Speech** to convert legal summaries into natural speech.
- Built secure authentication using **JWT** and implemented **notes management** and **history tracking** for case retrieval.

SteelDefectNet – Steel Surface Defect Detection Using Deep Learning

Oct 2024 – Apr 2025

Tech Stack: Python, YOLOv11, CNN, ResNet, VGG16, Vision Transformer, Roboflow, Flask

- Developed deep learning models including **YOLOv11**, **CNN**, **ResNet**, **VGG16**, and **Vision Transformer** to detect steel surface defects.
- Prepared and annotated a steel defect dataset using **Roboflow**, applied **data augmentation**, and evaluated performance using **precision**, **recall**, and **mAP**.
- Built a **Flask web interface** for uploading steel images and visualizing real-time defect predictions.

INTERNSHIP EXPERIENCE

AI/ML Intern – RGreen Technologies, Madurai

Jun 6 2025 – Jun 22 2025

- Developed a **chatbot system** with API integration and backend service development.
- Assisted in integrating **Machine Learning models** into the chatbot workflow and improved response handling.

App Development Intern – Micro Infotech, Coimbatore

Dec 8 2025 – Dec 23 2025

- Built an **Intelligent Quotation Creation and Management Application** for automated quotation generation.
- Implemented **UI components**, backend business logic, and secure data management.

Web Development Intern – Infotact Solutions

May 25 2025 – Aug 25 2025

- **Admin registers artisans** who upload handmade products for **platform review**.
- **Admin approves products** for dashboard display, enabling users to **browse and purchase**.

CO-CURRICULAR ACTIVITIES

Technical Activities

- Developed an **Online Leave Management System** during a Domestic Hackathon at SREC.
- Designed and deployed an **Internship Management Portal** for internship workflow management at SREC.
- Completed the **CIRCOR Industry Project** for automated **O-Ring detection** using **Mask R-CNN**.

Leadership and Achievements

- Ambassador – **AI Vishwaguru Initiative**, SREC.
- Executive Member – **Association of Master's Tech CSE**, SREC.
- Winner – **GitSetGo Contest (3rd Place)**, SREC.
- Runner-Up – **NiCHE Non-Technical Event**, organized by MBA Department at SREC.
- Event Coordinator – **Codeathon** during **UTSAVA 2026** (Techno-Cultural Fest), SREC.

CERTIFICATIONS

- **Python Foundation Certification** – Infosys Springboard
- **Generative AI: Hands-on Approach** – Sri Ramakrishna Engineering College
- **TCS iON Career Edge – Young Professional** – TCS iON
- **MongoDB University Skill Badges and Certifications** – MongoDB Atlas
- **Privacy and Security in Online Social Media** – NPTEL